

► PATHOLOGY—INFLAMMATION

Inflammation Response to eliminate initial cause of cell injury, to remove necrotic cells resulting from the original insult, and to initiate tissue repair. Divided into acute and chronic. The inflammatory response itself can be harmful to the host if the reaction is excessive (eg, septic shock), prolonged (eg, persistent infections such as TB), or inappropriate (eg, autoimmune diseases such as SLE).

SIGN	MECHANISM
Cardinal signs	
Rubor and calor	Redness and warmth. Vasodilation (relaxation of arteriolar smooth muscle) → ↑ blood flow. Mediated by histamine, prostaglandins, bradykinin, NO.
Tumor	Swelling. Endothelial contraction/disruption (eg, from tissue damage) → ↑ vascular permeability → leakage of protein-rich fluid from postcapillary venules into interstitial space (exudate) → ↑ interstitial oncotic pressure. Endothelial cell contraction is mediated by leukotrienes (C ₄ , D ₄ , E ₄), histamine, serotonin, bradykinin.
Dolor	Pain. Sensitization of sensory nerve endings. Mediated by bradykinin, PGE ₂ , histamine.
Functio laesa	Loss of function. Inflammation impairs function (eg, inability to make fist due to hand cellulitis).
Systemic manifestations (acute-phase reaction)	
Fever	Pyrogens (eg, LPS) induce macrophages to release IL-1 and TNF → ↑ COX activity in perivascular cells of anterior hypothalamus → ↑ PGE ₂ → ↑ temperature set point.
Leukocytosis	↑ WBC count; type of predominant cell depends on inciting agent or injury (eg, bacteria → ↑ neutrophils).
↑ plasma acute-phase reactants	Serum concentrations significantly change in response to acute and chronic inflammation. Produced by liver. Notably induced by IL-6.

Acute phase reactants

POSITIVE (UPREGULATED)	
C-reactive protein	Opsonin; fixes complement and facilitates phagocytosis. Measured clinically as a nonspecific sign of ongoing inflammation.
Ferritin	Binds and sequesters iron to inhibit microbial iron scavenging.
Fibrinogen	Coagulation factor; promotes endothelial repair; correlates with ESR.
Haptoglobin	Binds extracellular hemoglobin, protects against oxidative stress.
Hepcidin	↓ iron absorption (by degrading ferroportin) and ↓ iron release (from macrophages) → anemia of chronic disease.
Procalcitonin	Increases in bacterial infections.
Serum amyloid A	Prolonged elevation can lead to secondary amyloidosis.
NEGATIVE (DOWNREGULATED)	
Albumin	Reduction conserves amino acids for positive reactants.
Transferrin	Internalized by macrophages to sequester iron.
Transthyretin	Also called prealbumin. Reduction conserves amino acids for positive reactants.

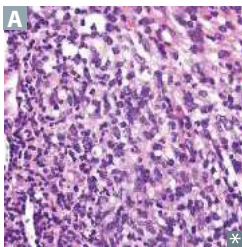
Erythrocyte sedimentation rate

RBCs normally remain separated via $\ominus$ charges. Products of inflammation (eg, fibrinogen) coat RBCs $\rightarrow$ $\downarrow$ $\ominus$ charge $\rightarrow$ $\uparrow$ RBC aggregation. Denser RBC aggregates fall at a faster rate within a pipette tube $\rightarrow$ $\uparrow$ ESR. Often co-tested with CRP (more specific marker of inflammation).

$\uparrow$ ESR	$\downarrow$ ESR ^a
Most anemias	Sickle cell anemia (altered shape)
Infections	Polycythemia ($\uparrow$ RBCs “dilute” aggregation factors)
Inflammation (eg, giant cell [temporal] arteritis, polymyalgia rheumatica)	HF
Cancer (eg, metastases, multiple myeloma)	Microcytosis
Renal disease (end-stage or nephrotic syndrome)	Hypofibrinogenemia
Pregnancy	

^aLower than expected.

Acute inflammation



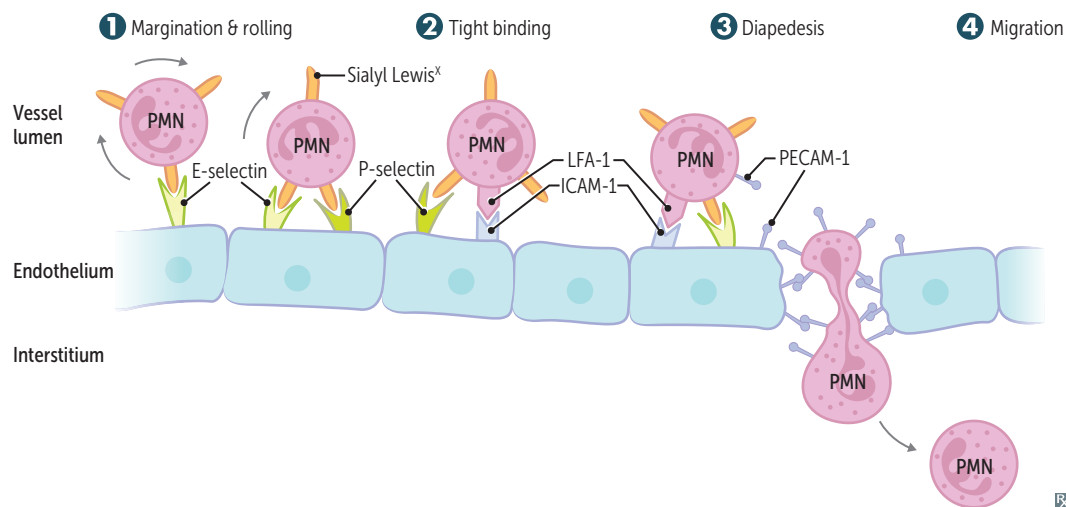
Transient and early response to injury or infection. Characterized by neutrophils in tissue **A**, often with associated edema. Rapid onset (seconds to minutes) and short duration (minutes to days). Represents a reaction of the innate immune system (ie, less specific response than chronic inflammation).

STIMULI	Infections, trauma, necrosis, foreign bodies.	
MEDIATORS	Toll-like receptors, arachidonic acid metabolites, neutrophils, eosinophils, antibodies (pre-existing), mast cells, basophils, complement, Hageman factor (factor XII).	Inflammasome —Cytoplasmic protein complex that recognizes products of dead cells, microbial products, and crystals (eg, uric acid crystals) $\rightarrow$ activation of IL-1 and inflammatory response.
COMPONENTS	▪ Vascular: vasodilation ($\rightarrow$ $\uparrow$ blood flow and stasis) and $\uparrow$ endothelial permeability (contraction of endothelial cells opens interendothelial junctions) ▪ Cellular: extravasation of leukocytes (mainly neutrophils) from postcapillary venules $\rightarrow$ accumulation of leukocytes in focus of injury $\rightarrow$ leukocyte activation	To bring cells and proteins to site of injury or infection. Leukocyte extravasation has 4 steps: margination and rolling, adhesion, transmigration, and migration (chemoattraction).
OUTCOMES	▪ Resolution and healing (IL-10, TGF-β) ▪ Persistent acute inflammation (IL-8) ▪ Abscess (acute inflammation walled off by fibrosis) ▪ Chronic inflammation (antigen presentation by macrophages and other APCs $\rightarrow$ activation of CD4+ Th cells) ▪ Scarring	Macrophages predominate in the late stages of acute inflammation (peak 2–3 days after onset) and influence outcome by secreting cytokines.

Leukocyte
extravasation

Extravasation predominantly occurs at postcapillary venules.

STEP	VASCULATURE/STROMA	LEUKOCYTE
1 Margination and rolling— defective in leukocyte adhesion deficiency type 2 (↓ Sialyl Lewis ^X)	E-selectin (upregulated by TNF and IL-1)	Sialyl Lewis ^X
	P-selectin (translocated to endothelial surface via exocytosis from Weibel-Palade bodies)	Sialyl Lewis ^X
	GlyCAM-1, CD34	L-selectin
2 Tight binding (adhesion)— defective in leukocyte adhesion deficiency type 1 (↓ CD18 integrin subunit)	ICAM-1 (CD54)	CD11/18 integrins (LFA-1, Mac-1)
	VCAM-1 (CD106)	VLA-4 integrin
3 Dia PE desis (transmigration)— WBC travels between endothelial cells and exits blood vessel	PECAM-1 (CD31)	PECAM-1 (CD31)
4 Migration—WBC travels through interstitium to site of injury or infection guided by chemotactic signals	Chemotactic factors: C5a, IL-8, LTB ₄ , 5-HETE, kallikrein, platelet-activating factor, N-formylmethionyl peptides	Various



Chronic inflammation Prolonged inflammation characterized by mononuclear infiltration (macrophages, lymphocytes, plasma cells), which leads to simultaneous tissue destruction and repair (including angiogenesis and fibrosis). May be preceded by acute inflammation.

STIMULI	Persistent infections (eg, TB, <i>T pallidum</i> , certain fungi and viruses) → type IV hypersensitivity, autoimmune diseases, prolonged exposure to toxic agents (eg, silica) and foreign material.
MEDIATORS	Macrophages are the dominant cells. Interaction of macrophages and T cells → chronic inflammation. - Th1 cells secrete IFN-γ → macrophage classical activation (proinflammatory) - Th2 cells secrete IL-4 and IL-13 → macrophage alternative activation (repair and anti-inflammatory)
OUTCOMES	Scarring, amyloidosis, and neoplastic transformation (eg, chronic HCV infection → chronic inflammation → hepatocellular carcinoma; <i>Helicobacter pylori</i> infection → chronic gastritis → gastric adenocarcinoma).

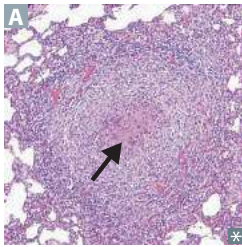
Wound healing

Tissue mediators	MEDIATOR	ROLE
	FGF	Stimulates angiogenesis
	TGF- β	Angiogenesis, fibrosis
	VEGF	Stimulates angiogenesis
	PDGF	Secreted by activated platelets and macrophages Induces vascular remodeling and smooth muscle cell migration Stimulates fibroblast growth for collagen synthesis
	Metalloproteinases	Tissue remodeling
	EGF	Stimulates cell growth via tyrosine kinases (eg, EGFR/ <i>ErbB1</i>)
PHASE OF WOUND HEALING	EFFECTOR CELLS	CHARACTERISTICS
Inflammatory (up to 3 days after wound)	Platelets, neutrophils, macrophages	Clot formation, ↑ vessel permeability and neutrophil migration into tissue; macrophages clear debris 2 days later
Proliferative (day 3–weeks after wound)	Fibroblasts, myofibroblasts, endothelial cells, keratinocytes, macrophages	Deposition of granulation tissue and type III collagen, angiogenesis, epithelial cell proliferation, dissolution of clot, and wound contraction (mediated by myofibroblasts) Delayed second phase of wound healing in vitamin C and copper deficiency
Remodeling (1 week–6+ months after wound)	Fibroblasts	Type III collagen replaced by type I collagen, ↑ tensile strength of tissue Collagenases (require zinc to function) break down type III collagen Zinc deficiency → delayed wound healing

Granulomatous inflammation

A pattern of chronic inflammation. Can be induced by persistent T-cell response to certain infections (eg, TB), immune-mediated diseases, and foreign bodies. Granulomas “wall off” a resistant stimulus without completely eradicating or degrading it → persistent inflammation → fibrosis, organ damage.

HISTOLOGY



Focus of epithelioid cells (activated macrophages with abundant pink cytoplasm) surrounded by lymphocytes and multinucleated giant cells (formed by fusion of several activated macrophages).
Two types:

Caseating: associated with **central necrosis** **A**. Seen with infectious etiologies (eg, TB, fungal).
Noncaseating: no central necrosis. Seen with noninfectious etiologies (eg, sarcoidosis, Crohn disease).

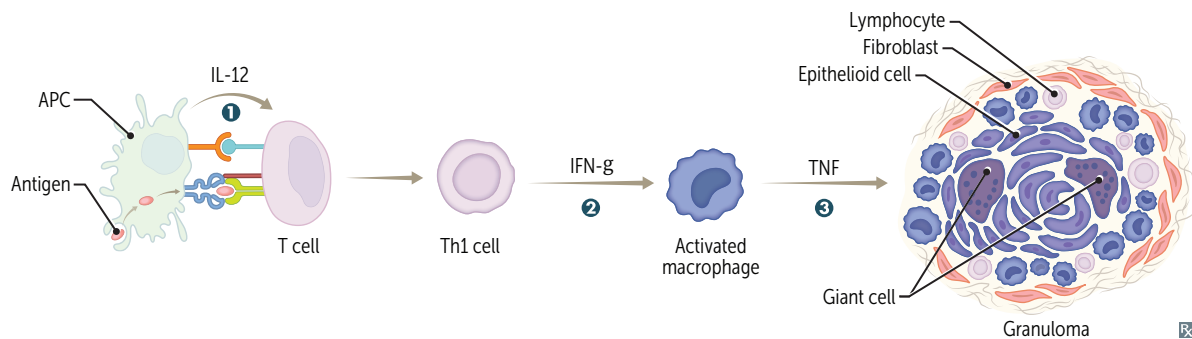
MECHANISM

- ① APCs present antigens to CD4+ Th cells and secrete IL-12 → CD4+ Th cells differentiate into Th1 cells
- ② Th1 secretes IFN- γ → macrophage activation
- ③ Macrophages ↑ cytokine secretion (eg, TNF) → formation of epithelioid macrophages and giant cells

Anti-TNF therapy can cause sequestering granulomas to break down → disseminated disease.

Always test for latent TB before starting anti-TNF therapy.

Associated with hypercalcemia due to ↑ 1 α -hydroxylase activity in activated macrophages, resulting in ↑ vitamin D activity.



ETIOLOGIES

Infectious

Bacterial: *Mycobacteria* (tuberculosis, leprosy), *Bartonella henselae* (cat scratch disease; stellate necrotizing granulomas), *Listeria monocytogenes* (granulomatosis infantiseptica), *Treponema pallidum* (3° syphilis)
Fungal: endemic mycoses (eg, histoplasmosis)
Parasitic: schistosomiasis
Catalase ⊕ organisms in chronic granulomatous disease

Noninfectious

Immune-mediated: sarcoidosis, Crohn disease, 1° biliary cholangitis, subacute (de Quervain/ granulomatous) thyroiditis
Vasculitis: granulomatosis with polyangiitis, eosinophilic granulomatosis with polyangiitis, giant cell (temporal) arteritis, Takayasu arteritis
Foreign bodies: berylliosis, talcosis, hypersensitivity pneumonitis

Scar formation

Occurs when repair cannot be accomplished by cell regeneration alone. Nonregenerated cells (2° to severe acute or chronic injury) are replaced by connective tissue. 70–80% of tensile strength regained at 3 months; little tensile strength regained thereafter. Excess TGF- β is associated with aberrant scarring, such as hypertrophic and keloid scars.

	Hypertrophic scar A	Keloid scar B
COLLAGEN SYNTHESIS	↑ (type III collagen)	↑↑↑ (types I and III collagen)
COLLAGEN ORGANIZATION	Parallel	Disorganized
EXTENT OF SCAR	Confined to borders of original wound	Extends beyond borders of original wound with “clawlike” projections typically on earlobes, face, upper extremities
RECURRENCE	Infrequent	Frequent
PREDISPOSITION	None	↑ incidence in people with darker skin

